REAL TIME

DEEPFAKE DETECTOR



THE DEFENDERS

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WHAT IS THE PROBLEM?



Deepfake technology is rapidly advancing, making it increasingly difficult to distinguish between real and fake videos. Cyber criminals are using deepfakes for scams, misinformation, identity theft, and even ransom demands, where fake videos are used to manipulate victims. Existing detection methods struggle with real-time analysis, accuracy, and accessibility—we aim to solve that.

Example: In 2023, a Hong Kong company was scammed out of $25 million because employees followed instructions from a deepfake video impersonating their CFO!

WHAT IS THE BIG DEAL?



01 MISINFORMATION & FAKE NEWS

Deepfakes are used to spread false information, influencing public opinion and elections.

02 FINANCIAL & RANSOM SCAMS

Criminals use deepfake videos to scam people and businesses, like impersonating CEOs for fraudulent transactions.

03 IDENTITY THEFT & PRIVACY VIOLATIONS

Attackers can use deepfakes to create fake accounts, impersonate individuals, or fabricate evidence.

04 NATIONAL SECURITY RISKS

Intelligence agencies warn that deepfake technology could be used for espionage or political manipulation.

» WHAT DO WE BRING TO THE TABLE?



✓ **Real-time deepfake detection**: not just post-processing.

✓ **Works on video calls**

✓**Detects multiple manipulation techniques:** face-swaps, lip-sync forgeries

✓ Can be integrated into mobile & web applications.

TECH STACK

FRONTEND

HTML, CSS,
JAVASCRIPT

BACKEND

NODEMON, SOCKETS,
EXPRESS

LIBRARIES

PYTORCH, NUMPY, FLASK, WEB RTC



» HOW IT WORKS?



01.

Video Input:

The user uses our technology during call.

02.

Frame Extraction

The video is broken down into frames.

03.

Feature Analysis

The AI model analyzes facial movements, lighting inconsistencies, and artifacts.

04.

Deepfake Classification

Our deep learning model predicts whether the video is real or fake

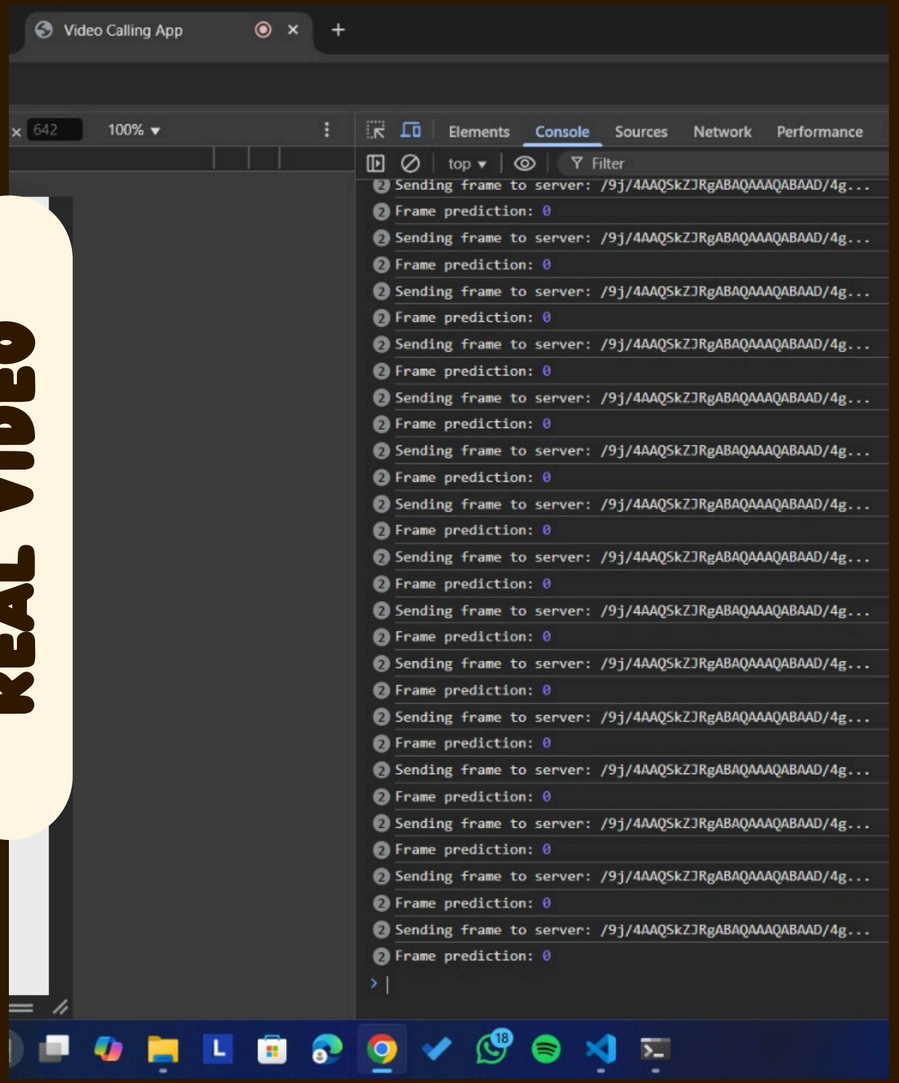
05.

Result

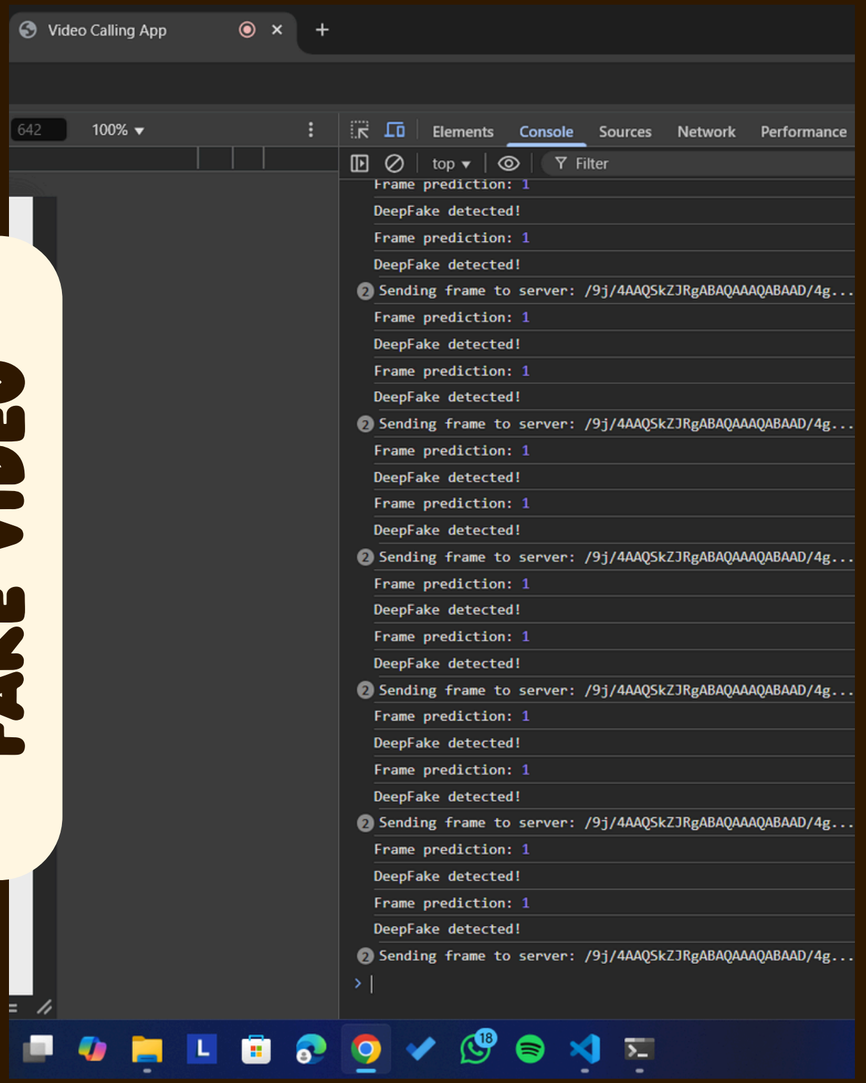
The result is displayed with a confidence score indicating the probability of manipulation.

RESULTS

REAL VIDEO



FAKE VIDEO





PERFORMANCE METRICS

ACCURACY

98.33%

PRECISION

98.97%

F1 SCORE

99.70

DATASET

DFD-MIT

ALGORITHM

CNN + VIT





FUTURE SCOPE

1. **Increase the dataset.**
2. **Integrate it in apps:** like Whatsapp, Zoom etc.
3. **Develop an API.**

Deepfakes can fake reality, but the truth will always need defenders.

THANK YOU

